

Sir Richard Arkwright and His Patent Granted in 1769

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SIR RICHARD ARKWRIGHT AND HIS PATENT
GRANTED IN 1769

By RICHARD L. HILLS
Director of the Manchester Museum of Science and Technology

[Plates 23 and 24]

FOR a person to rise from being a poor barber in a small town to a wealthy Knight of the Realm and Lord Lieutenant of the County of Derby suggests that he must have possessed some ability, even if it was only using other people's ideas. Sir Richard Arkwright's rise to riches and fame started with his spinning patent taken out in 1769. His fortune aroused considerable envy and hostility at the time, partly because people thought he had copied the inventions of others and partly because his patent held the key to the problem of spinning cotton by power-driven machinery and indeed his invention still forms the basis of the cotton-spinning industry today.

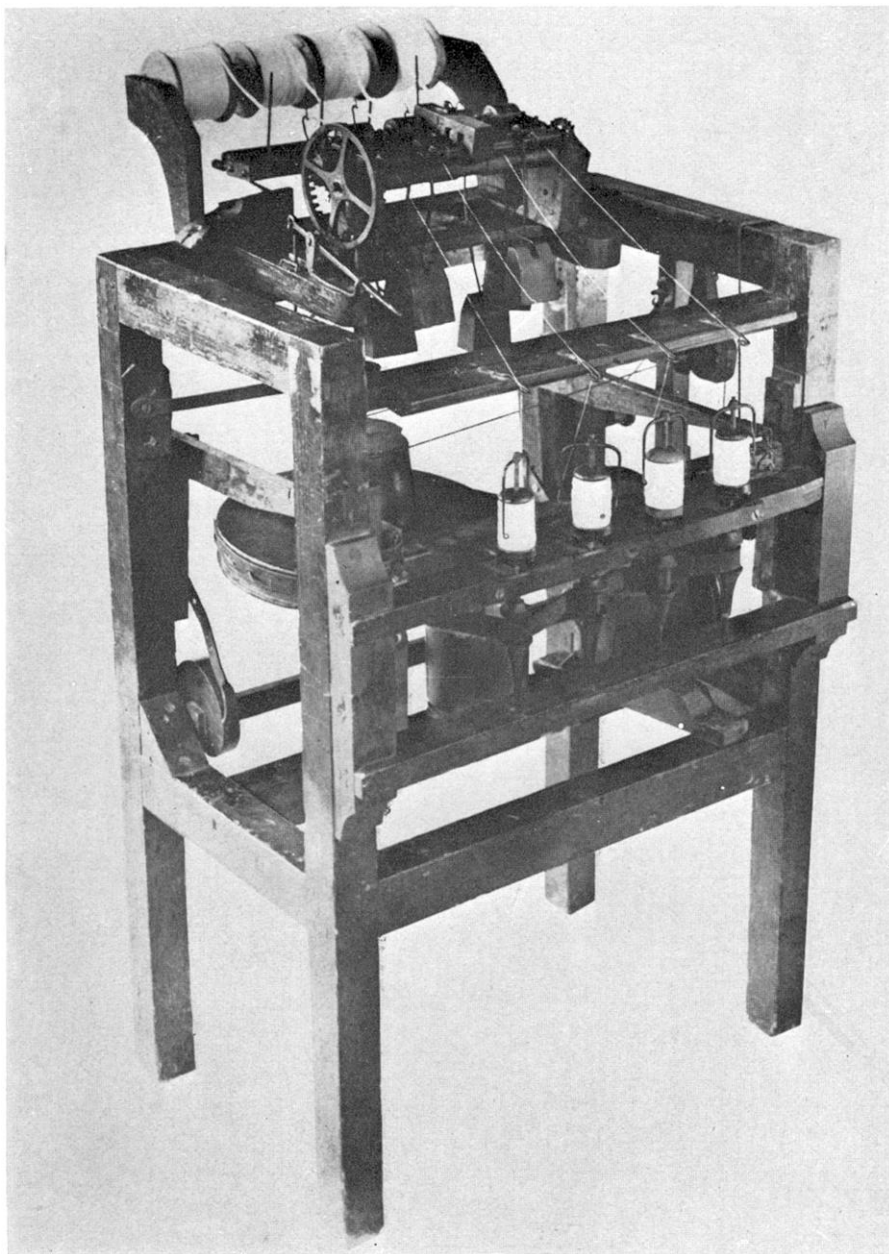
His simple, yet remarkable, invention of using pairs of rollers to replace human fingers opened a new chapter in industrial and human history, 'for Arkwright's cotton factories were examples of a new industrial organisation which gave rise to the factory system based on the use, first, of water power and later of the steam power that was soon to revolutionise it' (1). The English cotton industry already was beginning to oust Indian calicos from the African trade, and this process was hastened as cheaper cloth, made with 'twist' from the Arkwright water-powered mills, began to be produced in large quantities (2). This expanding export market encouraged the growth of mills at home, so that more mills, more power, waterwheels, steam engines, spinning machines, machinery manufacturers, machine-tool manufacturers were needed in ever-increasing numbers (2a). Greater experience in designing machines cheapened production and boosted sales, so that the cotton industry expanded rapidly in spite of severe slumps. Yet seldom can the author of such an important invention have met with so much hostility during his life or been the centre of so much debate after his death. This was partly through jealousy of his fortune, and partly because the originality of his invention has been doubted.

The earliest claimants to the idea of 'spinning by rollers' are Lewis Paul and John Wyatt who took out two patents in Paul's name in 1738 and 1758 (3). The variety of ideas included in their first patent suggests a machine



SIR RICHARD ARKWRIGHT (1732-1792)
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Arkwright's 'Water-Frame' from his mill at Matlock Bath

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still in the experimental stage. Correspondence and notes in the Wyatt manuscripts at Birmingham Public Library show that development continued until about 1743. The second patent merely describes this developed machine on which little work was done between 1743 and 1758, and Paul died soon after. Various mills were established with these machines, but none lasted long (4). Yet their machine was tantalizingly near to success, and might well have been successful for 'doubling' a thread, but it was based upon the wrong principles for spinning cotton.

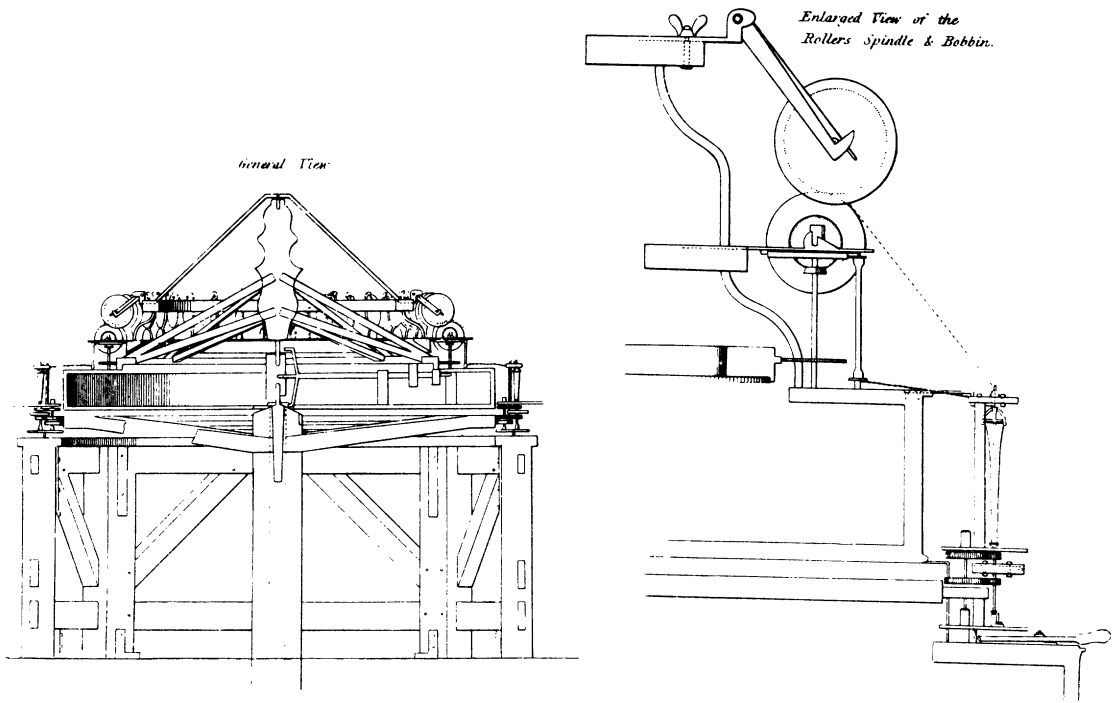
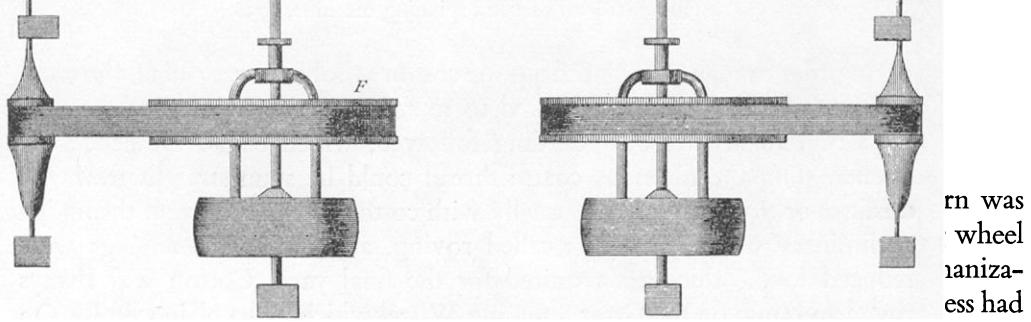


FIGURE 1. Paul's second spinning machine, 1758

In order to make a thread from the cotton wool in its raw state, the wool had first to be cleaned and picked to remove any impurities such as dirt, seeds or remains of pods. Carding followed, which began the process of disentangling the fibres. A coarse thread could be spun straight from the cardings or slivers, but more usually with cotton, the slivers went through a preliminary spinning process called roving, and then these rovings were reduced to the thickness required for the final yarn. Cotton was always roved and spun on the Great Spinning Wheel which had a plain spindle. On this type of wheel, the spinner first of all spun a length and then wound it



been solved.

Paul and Wyatt used the bobbin and flyer from the flax wheel and tried rollers for drawing out the rovings. Whatever ideas they may have had about using successive pairs of rollers in their first patent, these were abandoned in their later experiments, probably because the rollers were too large in diameter. In their later machines, they used the principle of 'spindle

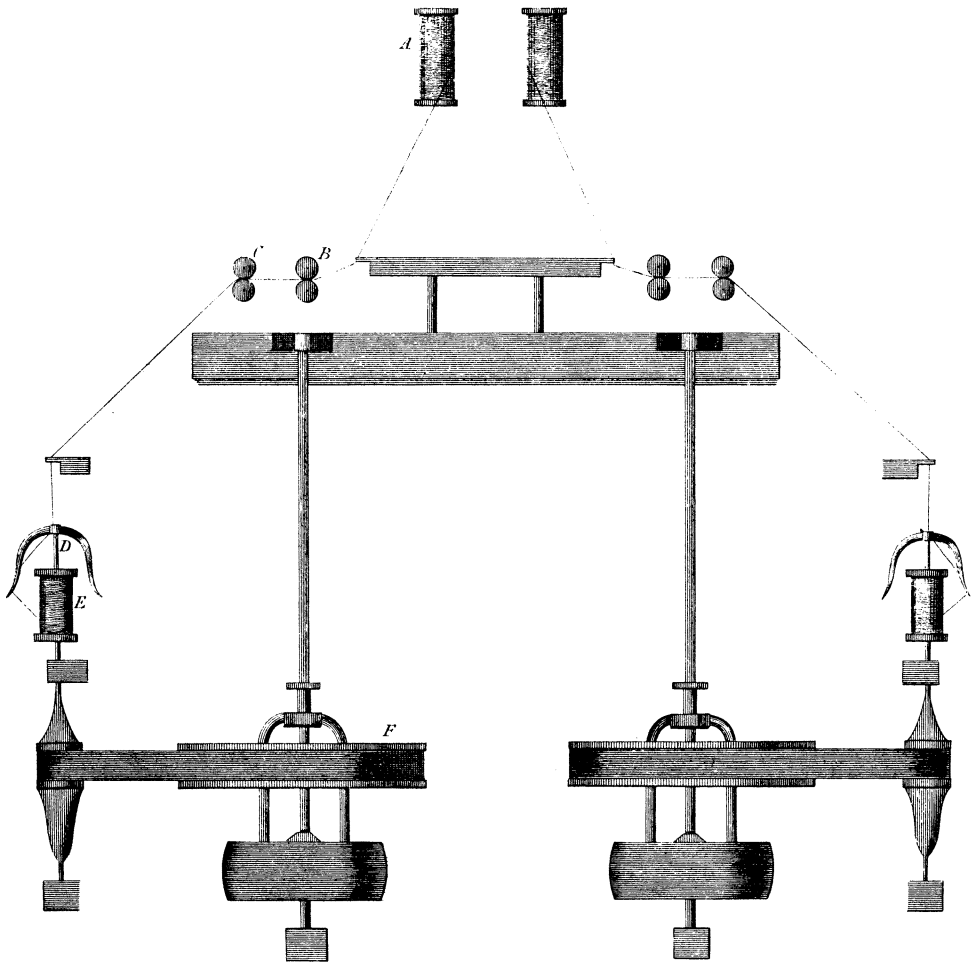


FIGURE 2. Drawing of High's water-frame as depicted by R. Guest

drawing', where the cotton was stretched to draw it out between the rollers and the spinning spindle. The roving was wound round one roller or cylinder which rested on the top of another, and was paid out between the rollers as they revolved. No attempt was made to use successive pairs of rollers to draw out the fibres, so that the cotton would have been drawn out unevenly and unevenly twisted, with many breaks.

It was some time before anyone else embarked on machine spinning after the failure of Paul and Wyatt. Around 1763, James Hargreaves developed his Jenny based on the intermittent principle of the Great Wheel, but this machine could spin only soft coarse yarns (5). About the same time, Thomas Highs of Leigh was experimenting with pairs of rollers, each set moving faster than the last, for drawing out the cotton. Somehow Richard Arkwright heard about this invention and employed John Kay, a Warrington clockmaker, to make a model of it. The exact relationship between these three has never been, and probably never will be, properly established, but the drawing of Highs's machine by his chief protagonist, Richard Guest, suggests that Highs failed in two vital points (6). First, he spaced his rollers too far apart so that the cotton would not have drawn out evenly. The distance between the rollers must be just greater than the staple length of the individual fibres of cotton. If the rollers are set too close together, two pairs of rollers grip the same cotton fibres which are broken by the pull of the faster moving rollers in front. If the rollers are set too far apart, the fibres float between them in the drafting zone so control is lost and an uneven yarn or breakages is the result. Secondly, Highs does not appear to have weighted his rollers, so that not only would they have failed to grip the cotton properly, but the twist imparted by the spindle would have tended to run up the yarn, between the last pair of rollers, and to twist the fibres in the drafting zone so that they would not have drawn out properly.

While it is realized that these reasons for Highs's failure based on Guest's evidence are highly conjectural, there is no doubt that Richard Arkwright persevered and was granted a patent on 15 July 1769 (7), for a properly developed spinning machine with four pairs of rollers, each pair moving proportionately faster than the last, set at the correct distance apart from each other for spinning cotton, and also weighted to secure adequate control. In the patent specification, Arkwright claimed that he 'had by great study and long application invented a new Piece of Machinery never before found out, practised or used for the making of Weft or Yarn from Cotton, Flax, and Wool, which would be of great Utility to a great many Manufacturers in His Kingdom of England, as well as to his Subjects in general, by Employing a great Number of Poor People in Working the said Machinery, and by

Making the said Weft or Yarn much Superior in Quality to any ever heretofore Manufactured or Made' (8). Although the machine depicted in the patent would not have spun flax or wool, Arkwright was able to develop it successfully for spinning cotton. In the patent, he envisaged horses providing the motive power, but soon something on a much larger scale was necessary.

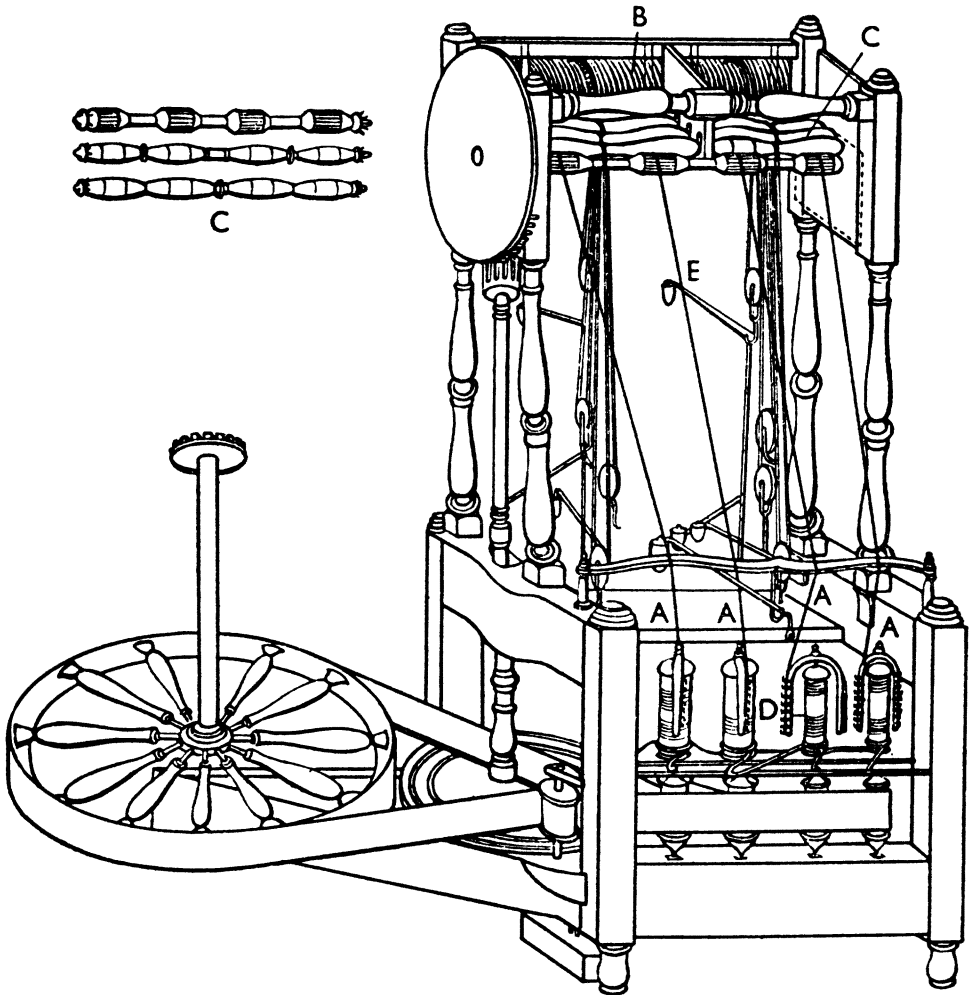


FIGURE 3. Drawing of Arkwright's spinning machine from the 1769 patent

Some of the stages in his path to success have been revealed in two documents recently presented to the University of Manchester Institute of Science and Technology. Although the patent was entered under the name of Arkwright himself, he was financed by two others, John Smalley and David Thornley, who came from Preston. They signed Articles of Agreement on 19 June 1769 to become Joint Proprietors and Partners, to share the profits and to take out a patent. They were to

Advance in equal Proportions all such sums of money as might be necessary in applying for a Patent as aforesaid and for Improving Enlarging and Working the Machine already Invented and others to be constructed for the same or the like purposes and all incident Charges and Expenses that might attend the same and the Carrying on such branches of Trade by Virtue thereof as might be thought most Advantageous (9).

The connexion of Smalley and Thornley with Preston may indicate that Arkwright had not left Lancashire when he took out his patent and that it was a little later when he set up some of his machines worked by horse-power in a mill at Nottingham. Nottingham promised a profitable market for their yarn because it was the centre of the hosiery business, but the venture proved beyond the resources of the three original partners. For greater financial support, the partnership was enlarged. Richard Arkwright, John Smalley, Samuel Need, Jedediah Strutt and David Thornley, 'by a certain indenture or Deed of Partnership Date the twelfth day of June in the year of our Lord one thousand seven hundred and seventy' agreed that they would become 'Partners, Joint Traders and Dealers together in the aforesaid particular Branch of Business Trade or New Invention . . . in the Manufacturing and Making Weft or Yarn from Cotton Flax and Wool . . . and should also become Owners and Proprietors of certain Leasehold Premises in Nottingham' (10).

Although there was a further period of trial, and much greater expenditure had to be incurred before the venture proved to be a financial and technical success, the results must have been promising enough for the partners to expand their operations. David Thornley had died before August 1771 when the other partners leased the site at Cromford and 'All that River Stream or Brook called Bonsall Brook situated and being within the Liberty of Cromford aforesaid together with the stream of water Issuing and running from Cromford Sough' (11). Here they were given authority for building mills for 'Spinning, Winding, or Throwing Silk, Worsted, Linen, Cotton or other materials' (12), but they seem to have

concentrated on cotton and established the first successful water-powered cotton-spinning mill.

The steady supply of water from Cromford Sough, together with the Bonsall Brook, must have seemed more than adequate to power their continuing spinning experiments, for a letter from Arkwright to Strutt in 1772 (13) shows they were still having difficulties. Yet within a decade, textile mills grew so quickly that the Cromford mill was found to be too small. From probably three or four horses at Nottingham and probably ten horse-power at Cromford, by the turn of the century, mills with water-wheels of over 100 horse-power had been built. The revolution continued with the adaptation of the rotative steam engine, and 'the importance of the invention may be appreciated if one remembers that today, two hundred years later, with all the improvements that have taken place in spinning machinery, the roller system is still the basis of the drawing process' (14).

NOTES

- (1) Hill, D. W. *Richard Arkwright and the Water Frame*. A Lecture given at the Watt-Arkwright Bicentenary Celebrations. The Institution of Mechanical Engineers, London, 1969.
- (2) Edwards, M. M. *The Growth of the British Cotton Trade, 1780-1815*. Manchester, 1967.
- (2a) This is more fully discussed in *Power in the industrial revolution* by R. L. Hills, to be published by Manchester University Press, February 1970.
- (3) Patent 562, 1738, and Patent 724, 1758.
- (4) Wadsworth, A. P. & Mann, J. de L. *The Cotton Trade and Industrial Lancashire*. Manchester, 1931.
- (5) Aspin, C. & Chapman, S. D. *James Hargreaves and the Spinning Jenny*. Preston, 1964.
- (6) Guest, R. *A Compendious History of the Cotton-Manufacture: with a Disproval of the Claim of Sir Richard Arkwright to the Invention of its ingenious Machinery*. Manchester, 1823.
- (7) Patent 931, 1769.
- (8) *Ibid.*
- (9) Articles of Agreement between Richard Arkwright, John Smalley and David Thornley, 19 June 1769.
- (10) Assignment of Mr Smalley's Fifth of the Nottingham and Cromford Spinning Works and the Patent, 1 October 1777.
- (11) *Ibid.*
- (12) *Ibid.*
- (13) Letter of R. Arkwright to J. Strutt, 2 March 1772. Birmingham Public Library.
- (14) Hill, D. W., *op. cit.*